

Quantization as Projection Kernel: Why Classical and Quantum Are Related by a Kernel

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Abstract

Every known quantization scheme — canonical (Dirac), geometric (Kostant–Souriau), Berezin–Toeplitz, deformation (Kontsevich), BRST, Weyl–Wigner, path integral (Feynman) — maps a commutative classical algebra to a non-commutative quantum algebra via a structure-preserving map with three universal invariants: (i) bracket preservation on an admissible subalgebra (Dirac’s rule), (ii) non-trivial null space (Groenewold–van Hove obstruction), and (iii) image/null decomposition of the classical observable algebra. We identify this as the kernel pattern formalized in Volumes I–X of the Projected Ontology Theory (POT) series. Different quantization schemes realize the image/null decomposition through different algebraic mechanisms: Berezin–Toeplitz via an idempotent Bergman projection ($P^2 = P$), BRST via a nilpotent cohomological operator ($Q^2 = 0$), and deformation quantization via a continuous star-product deformation parameterized by $\hbar$. What unifies them is not idempotence but the decomposition itself: every scheme partitions the classical algebra into a part that maps faithfully to the quantum algebra (image) and a part that does not (null space). We prove that the Groenewold obstruction is not a defect of quantization but a structural necessity: the null space is non-empty by construction, and the discrepancy between classical and quantum is controlled by $\hbar$. As a worked example, we apply the quantization kernel to the harmonic oscillator and show that: the discrete spectrum $E_n = \hbar\omega(n + 1/2)$ is the image; the zero-point energy $\hbar\omega/2$ is a property of the kernel with no classical analog; and the energy spacing $\hbar\omega$ is the sampling resolution. Volume X proved that infinities are null-space elements of the Hadamard projection; this volume proves that the quantum algebra itself is the image of a quantization kernel applied to the classical algebra. The ontological conclusion: classical and quantum mechanics are related by a structure-preserving kernel whose null space is the uncertainty principle. All 26 structural results are machine-verified by the Z3 SMT solver in the Kleis formal verification language.

Keywords: quantization, projection kernel, Poisson bracket, commutator, Dirac correspondence, Groenewold-van Hove theorem, Berezin-Toeplitz, BRST cohomology, deformation quantization, star product, Kontsevich, harmonic oscillator, projected ontology, formal verification, Z3

1 Introduction

The passage from classical mechanics to quantum mechanics is the oldest open structural question in theoretical physics. Dirac’s 1925 insight [Dir1925] identified the core operation: replace the Poisson bracket $\{f, g\}$ with the quantum commutator $[\hat{f}, \hat{g}]/(i\hbar)$, and require that this replace-

ment preserves the Lie algebra structure. The ‘quantization problem’ has since been attacked from many directions — geometric quantization (Kostant [Kos1970], Souriau [Sou1970]), Berezin–Toeplitz quantization (Berezin [Ber1975], Bordemann–Meinrenken–Schlichenmaier [BMS1994]), deformation quantization (Bayen et al. [BFFLS1978], Kontsevich [Kon1997]), BRST quantization (Becchi–Rouet–Stora [BRS1976], Tyutin [Tyu1975]), Weyl–Wigner quantization (Weyl [Wey1927], Wigner [Wig1932], Moyal [Moy1949]) — and in every case the resulting map has the same algebraic anatomy:

1. A **bracket-preserving** map on an admissible subalgebra (Dirac’s rule) from a commutative source to a non-commutative image.
2. A **non-trivial null space** (the Groenewold–van Hove obstruction [Gro1946, vH1951]): the kernel is genuinely lossy.
3. An **image/null decomposition** of the classical algebra into a part that quantizes faithfully and a part that does not.

Different schemes realize this decomposition through different algebraic mechanisms — idempotent projection (Berezin–Toeplitz), nilpotent cohomology (BRST), or continuous deformation (star product) — but all three invariants are present in every case. This paper argues that this pattern is not a coincidence. It is the kernel pattern formalized in Volumes I–X of the Projected Ontology Theory (POT) series. The same algebraic structure that identifies physical fields as images of Green’s function kernels (Volumes I–VI), that identifies renormalization as a projection kernel (Volume VII), and that identifies infinities as null-space elements of the Hadamard projection (Volume X) also identifies quantum mechanics as the image of a quantization kernel applied to classical mechanics.

The paper is organized as follows. Section 2 constructs the quantization kernel and establishes Dirac’s correspondence. Section 3 proves the Groenewold obstruction: the kernel has a non-trivial null space. Section 4 shows that Berezin–Toeplitz quantization is literally a projection. Section 5 constructs the BRST cohomological projection. Section 6 presents deformation quantization as a kernel deformation. Section 7 addresses the ‘overloaded projection’ objection by distinguishing the three algebraic realizations and identifying what actually unifies them. Section 8 applies the framework to the harmonic oscillator. Section 9 gives a structural remark on the analogy with sampling theory. Section 10 connects to Volumes VII and X. Section 11 draws the ontological conclusion.

2 The Quantization Kernel

We formalize the quantization map as a kernel in the POT sense.

The classical algebra is the space of smooth functions on a symplectic manifold (M, ω) , equipped with:

- Pointwise multiplication: $(fg)(x) = f(x)g(x)$, which is **commutative**.
- Poisson bracket: $\{f, g\} = \omega^{ij} \partial_i f \partial_j g$, which satisfies antisymmetry, the Jacobi identity, and the Leibniz rule.

The quantum algebra is the space of operators on a Hilbert space $\mathcal{H}$, equipped with:

- Operator multiplication: $(AB)|\psi\rangle = A(B|\psi\rangle)$, which is **non-commutative**.
- Commutator: $[A, B] = AB - BA$, which satisfies antisymmetry and the Jacobi identity.

The quantization kernel $Q : C^\infty(M) \rightarrow \mathcal{B}(\mathcal{H})$ maps classical observables to quantum operators. Dirac's canonical quantization rule (1925) requires:

$$Q(\{f, g\}) = \frac{1}{i\hbar}[Q(f), Q(g)]$$

for observables in the **admissible subalgebra**. For the fundamental pair (q, p) this gives:

$$Q(\{q, p\}) = Q(1) = \frac{1}{i\hbar}[\hat{q}, \hat{p}]$$

which forces $[\hat{q}, \hat{p}] = i\hbar$ — the canonical commutation relation. The bracket structure of classical mechanics is preserved in the image.

Theorem 1 (Classical commutativity). The source algebra is commutative: $fg = gf$ for all classical observables f, g .

Theorem 2 (Quantum non-commutativity). The image algebra is non-commutative: there exist operators A, B with $AB \neq BA$.

Both are axiomatized in `QuantizationKernel` and verified by Z3.

3 The Groenewold Obstruction: Non-Trivial Null Space

Can Dirac's rule hold for **all** classical observables? Groenewold [Gro1946] and van Hove [vH1951] proved the answer is no.

Theorem 3 (Groenewold–van Hove). There is no linear map $Q : C^\infty(M) \rightarrow \mathcal{B}(\mathcal{H})$ satisfying $Q(\{f, g\}) = (1/i\hbar)[Q(f), Q(g)]$ for all $f, g \in C^\infty(M)$.

Specifically, Groenewold showed that for the harmonic oscillator phase space $T^*\mathbb{R}$, the identity $\{q^2, p^2\} = 4qp$ combined with $Q(\{q^2, p^2\}) = (1/i\hbar)[Q(q^2), Q(p^2)]$ produces a contradiction with $Q(qp) = (\hat{q}\hat{p} + \hat{p}\hat{q})/2$ at the level of third-order polynomials. The obstruction is inherent: it cannot be removed by any choice of ordering prescription.

In the POT framework, this means the quantization kernel Q has a **non-trivial null space**: the set of classical observables for which bracket preservation fails. This is not a defect — it is the **structural content** of quantization. The null space is where classical and quantum mechanics genuinely disagree, and the discrepancy is controlled by $\hbar$.

The parallel with Volume X is precise:

Axiomatized in `QuantizationKernel` (Dirac + Groenewold) and verified by Z3.

4 Berezin–Toeplitz: Quantization Is Literally Projection

	Vol X (Hadamard)	Vol XI (Quantization)
Source	Formal series	Classical observables
Image	Finite observables	Quantum operators
Null space	UV divergences	Ordering ambiguities
Admissible domain	$n \geq 0$ coefficients	Linear/quadratic observables
Obstruction	Laurent poles	Groenewold–van Hove
Projection	Hadamard P	Quantization Q

Table 1: Structural parallel between Volumes X and XI.

On a compact Kähler manifold (M, ω, J) , the Berezin–Toeplitz quantization scheme makes the projection structure fully explicit. Let $L^{\otimes m}$ denote the m -th tensor power of the prequantum line bundle, and let $H^0(M, L^{\otimes m})$ be the space of holomorphic sections. The Bergman projection

$$P_m : L^2(M, L^{\otimes m}) \rightarrow H^0(M, L^{\otimes m})$$

is the orthogonal projection onto holomorphic sections. Its integral kernel is the Bergman kernel $K_{m(x,y)}$, from which the Toeplitz operator is constructed:

$$T_{m(f)} = P_m \circ M_f : H^0 \rightarrow H^0$$

where M_f is multiplication by the classical observable f . The quantum operator associated to f is $T_{m(f)}$.

The Bergman projection P_m satisfies the POT kernel axioms:

1. **Idempotent:** $P_m^2 = P_m$. Applying the projection twice is the same as applying it once.
2. **Image = quantum Hilbert space:** holomorphic sections $s \in H^0$ are fixed: $P_{m(s)} = s$.
3. **Non-trivial null space:** non-holomorphic sections are projected: $P_{m(s)} \neq s$ for $s \notin H^0$.

Bordemann, Meinrenken, and Schlichenmaier [BMS1994] proved that as $m \rightarrow \infty$:

$$T_{m(f)}T_{m(g)} - T_{m(fg)} = O(1/m)$$

$$m[T_{m(f)}, T_{m(g)}] - iT_{m(\{f,g\})} = O(1/m)$$

which establishes that Berezin–Toeplitz quantization recovers the Dirac correspondence in the semiclassical limit (with $\hbar \sim 1/m$).

The ontological reading: the Bergman kernel **is** the quantization kernel. The classical algebra lives on L^2 . The quantum algebra lives on H^0 . The passage from classical to quantum is orthogonal projection — literally the same mathematical operation as the Hadamard projection of Volume X, applied in a different space.

Axiomatized in **BerezinToeplitz** and verified by Z3 (4 results).

5 BRST Cohomology: Physical States as Cohomological Projection

In gauge quantum field theory, the full Fock space includes unphysical ghost degrees of freedom introduced by gauge fixing [FP1967]. The BRST operator Q (Becchi–Rouet–Stora [BRS1976], Tyutin [Tyu1975]) is a nilpotent operator acting on the extended Fock space:

$$Q^2 = 0$$

This nilpotency defines a cohomological projection. The physical Hilbert space is the BRST cohomology:

$$\mathcal{H}_{\text{phys}} = \text{Ker}(Q) / \text{Im}(Q)$$

The structure is:

- **BRST-closed states:** $Q|\psi\rangle = 0$ (candidates for physical states).
- **BRST-exact states:** $|\psi\rangle = Q|\chi\rangle$ (pure gauge, unphysical).
- **Physical states:** closed but not exact.

Because $Q^2 = 0$, every exact state is automatically closed ($\text{Im}(Q) \subseteq \text{Ker}(Q)$), so the quotient is well-defined. The image of the BRST projection is the physical Hilbert space; the null space (exact states) contains the gauge artifacts.

This is the POT kernel pattern applied to gauge theories:

- $\text{Image}(Q^{\text{BRST}}) = \text{physical states (gauge-invariant)}$.
- $\text{Null}(Q^{\text{BRST}}) = \text{gauge artifacts (ghosts, longitudinal photons)}$.

The BRST projection is not idempotent in the same sense as the Bergman projection (it is nilpotent, $Q^2 = 0$, rather than $Q^2 = Q$), but the **cohomological quotient** it defines is a genuine projection from the extended space to the physical subspace. The two kinds of projection — idempotent and cohomological — are unified in the POT framework as different realizations of the image/null decomposition.

Axiomatized in `BRSTProjection` and verified by Z3 (4 results).

6 Deformation Quantization: Star Product as Kernel Deformation

Deformation quantization (Bayen, Flato, Frønsdal, Lichnerowicz, and Sternheimer [BFFLS1978]) reformulates quantum mechanics directly on phase space, without Hilbert spaces or operators, by deforming the classical product:

$$f \star g = fg + \sum_{k=1}^{\infty} \hbar^k B_k(f, g)$$

where B_k are bidifferential operators (determined by the Poisson structure). The key property:

$$f \star g - g \star f = i\hbar\{f, g\} + O(\hbar^2)$$

Kontsevich [Kon1997] proved that a star product exists on **every** Poisson manifold (his formality theorem), settling a 20-year conjecture. The bidifferential operators B_k are themselves integral kernels — they define the quantization via convolution.

The star product demonstrates the kernel structure of quantization:

- At $\hbar = 0$: $f \star g = fg$ (commutative, classical).
- At $\hbar > 0$: $f \star g \neq g \star f$ in general (non-commutative, quantum).
- The deformation parameter $\hbar$ controls the ‘distance from classical.’

In the POT language: the star product is a one-parameter family of kernels $Q_\hbar$ interpolating between the identity kernel ($\hbar = 0$, classical) and the quantum kernel ($\hbar > 0$). The ‘quantization’ is the deformation itself; the classical limit is the trivial projection.

Theorem 4. The commutative limit of the star product recovers the classical product: if $f \star g = g \star f$ for all f, g , then $f \star g = fg$.

Axiomatized in `DeformationQuantization` and verified by Z3 (3 results).

7 Three Realizations of the Image/Null Decomposition

A legitimate objection: the three quantization schemes discussed in Sections 4–6 use different algebraic mechanisms. Are we simply calling everything a ‘projection’? We address this directly.

The three mechanisms are genuinely different:

1. **Idempotent** (Berezin–Toeplitz): $P^2 = P$. The Bergman projection is an orthogonal projection in the functional-analytic sense. Image and null space are orthogonal complements in L^2 .
2. **Cohomological** (BRST): $Q^2 = 0 \neq Q$. The BRST operator is nilpotent, not idempotent. The ‘projection’ is the passage to the cohomology $\text{Ker}(Q)/\text{Im}(Q)$, which is a quotient, not an orthogonal complement.
3. **Deformative** (star product): neither $P^2 = P$ nor $Q^2 = 0$. The star product is a continuous one-parameter deformation. The ‘projection’ is the limiting map $\hbar \rightarrow 0$ that recovers the classical product.

What unifies them is **not** a shared algebraic operation. It is a shared **structural pattern**: each scheme decomposes the classical observable algebra into two parts — a part that maps faithfully to the quantum algebra (the image) and a part where the classical-quantum correspondence breaks down (the null space). The decomposition satisfies, in every case:

1. The image algebra is non-commutative (quantum).
2. The bracket is preserved on the admissible part of the source (Dirac’s rule holds on a subalgebra).
3. The null space is non-trivial (Groenewold–van Hove obstruction).

We formalize this as follows. A **quantization kernel** is a linear map $Q : \mathcal{A}_{\text{cl}} \rightarrow \mathcal{A}_{\text{qu}}$ together with a decomposition $\mathcal{A}_{\text{cl}} = \mathcal{A}_{\text{adm}} \oplus \mathcal{A}_{\text{null}}$ such that:

- Q restricted to $\mathcal{A}_{\text{adm}}$ is a Lie algebra homomorphism (bracket-preserving).
- Q restricted to $\mathcal{A}_{\text{null}}$ is **not** a Lie algebra homomorphism.
- $\mathcal{A}_{\text{null}} \neq \{0\}$ (the obstruction is non-trivial).

This definition does not require idempotence. Berezin–Toeplitz, BRST, and deformation quantization are all instances. The contribution of this paper is not to claim they are ‘the same operation,’ but to identify that they share these three invariants — and that these invariants are the same ones that appear in the POT kernel pattern of Volumes I–X.

All three invariants are axiomatized and verified by Z3 in the theory file.

8 Worked Example: The Harmonic Oscillator

As a concrete end-to-end demonstration, we apply the quantization kernel to the harmonic oscillator — the simplest system that exhibits all three structural invariants.

8.0.1 The classical system

The classical Hamiltonian is:

$$H(q, p) = \frac{p^2}{2m} + \frac{m\omega^2 q^2}{2}$$

Phase-space orbits are ellipses of area $A(E) = 2\pi E/\omega$. The energy E is **continuous**: any non-negative real value is allowed. The fundamental Poisson bracket is $\{q, p\} = 1$.

8.0.2 The quantum system (image of the kernel)

The quantization kernel maps $H \rightarrow \hat{H}$ with eigenvalues:

$$E_n = \hbar\omega(n + 1/2), \quad n = 0, 1, 2, \dots$$

This is the **image** of Q applied to the classical Hamiltonian. The three structural invariants and their consequences:

1. Energy levels are discrete. $E_0 = 0.5, E_1 = 1.5, E_2 = 2.5, E_3 = 3.5, E_4 = 4.5$ (in natural units $\hbar = \omega = 1$). Verified by computation.

2. Zero-point energy is a quantum artifact. $E_0 = \hbar\omega/2 = 0.5 > 0$, while the classical minimum energy is $E = 0$ (oscillator at rest). The zero-point energy has **no classical analog** — it is a property of the quantization kernel, not the classical source. The uncertainty principle $\Delta q \Delta p \geq \hbar/2$ prevents the oscillator from occupying the phase-space origin.

3. Energy spacing is constant (sampling resolution). $\Delta E = E_{n+1} - E_n = \hbar\omega = 1.0$ for all n . This is the ‘sampling resolution’ of the quantization kernel: the minimum energy difference the quantum system can distinguish. Below this resolution, energy differences lie in the null space.

4. Dirac’s correspondence is preserved. The fundamental bracket maps: $\{q, p\} = 1 \rightarrow [\hat{q}, \hat{p}]/(i\hbar) = 1$. The kernel preserves the Lie algebra structure.

5. Phase-space area is quantized. $\Delta A = A(E_{n+1}) - A(E_n) = 2\pi\hbar = h$. The phase space is partitioned into cells of area h (Planck’s constant). The minimum cell area is $A(E_0) = \pi\hbar > 0$: the Nyquist limit of the quantization kernel.

8.0.3 The classical limit

As $n \rightarrow \infty$: $\Delta E/E_n = 1/(n + 1/2) \rightarrow 0$. At $n = 100$, the relative spacing is below 1% — the discrete spectrum is effectively continuous. The quantization kernel approaches the identity at large quantum numbers: classical mechanics is the limit of trivial projection.

All 7 numerical results are verified by direct computation in `theories/pot_quantization_kernel_worked.kleis`.

9 Structural Remark: Correspondence with Discrete-Time Control

We note a structural analogy — not a physical identification — between the quantization kernel and discretization in digital control theory.

In hybrid control systems, a continuous-time plant is controlled by a discrete-time controller through an analog-to-digital converter (ADC). The ADC samples the continuous signal at period T , producing a discrete-time representation. The correspondence is algebraic:

Both are instances of the same algebraic pattern: a linear, structure-preserving map from a continuous algebra to a discrete algebra, with a resolution parameter controlling the information loss and a fundamental limit on distinguishable states.

We emphasize that this is a **structural observation**, not a physical claim. We are not asserting that quantization ‘is’ sampling, or that $\hbar$ ‘is’ a sampling period. The observation is that the same structural pattern (bracket/transform preservation on an admissible domain, non-trivial null space, resolution limit) arises independently in two different engineering/physical contexts. This universality is evidence that the projection kernel pattern is a fundamental structural phenomenon, not an accident of quantum mechanics.

10 Connection to Volumes VII and X

The quantization kernel completes a three-volume arc within the POT series:

	Quantization	Digital Control
Continuous source	Phase space $C^\infty(M)$	Continuous plant $G(s)$
Discrete image	Hilbert space $\mathcal{H}$	Discrete model $G(z)$
Resolution parameter	$\hbar$	Sampling period T
Information loss	Uncertainty principle	Aliasing
Transform pair	Poisson $\rightarrow$ commutator	Laplace $\rightarrow$ Z-transform
Minimum cell	$\Delta q \Delta p \geq \hbar/2$	Nyquist: $f_{\max} < 1/(2T)$

Table 2: Structural analogy between quantization and discrete-time control.

	Vol VII	Vol X	Vol XI
Kernel	K_{QFT}	P (Hadamard)	Q (quantization)
Source	QFT source J	Formal series	Classical $C^\infty(M)$
Image	Finite observable	Admissible series	Quantum $\mathcal{B}(\mathcal{H})$
Null space	Divergent intermediate	UV poles / singularities	Ordering ambiguity
Admissible	Regularized values	$n \geq 0$ coefficients	Linear/quadratic obs.
Decomposition	$\text{FP}^2 = \text{FP}$	$P^2 = P$	Image/Null (Sec. 7)

Table 3: The POT kernel pattern across three volumes.

Volume VII: Renormalization as projection kernel. The composite QFT kernel $K_{\text{QFT}} = \text{FP} \circ K_{\text{ren}} \circ K_{\text{path}}$ maps sources to finite observables. The finite-part operator FP is a projection: it extracts admissible content and discards UV divergences.

Volume X: Infinities as null-space elements. The Hadamard projection P on formal power series decomposes every amplitude into $\text{Image}(P)$ (finite observables) and $\text{Null}(P)$ (UV divergences, curvature singularities). The infinities are representational artifacts.

Volume XI (this paper): Quantum mechanics as projection image. The quantization kernel Q maps classical observables to quantum operators. $\text{Image}(Q)$ is the quantum algebra. $\text{Null}(Q)$ is the Groenewold obstruction.

The pattern is identical across all three:

The unification is not metaphorical. The same structural pattern — bracket preservation on an admissible domain, non-trivial null space, image/null decomposition — applies in each case. The three volumes together establish that the POT kernel pattern governs: (1) how finite observables emerge from divergent intermediates (Vol VII), (2) why infinities are representational artifacts (Vol X), and (3) why the quantum world is a projected image of the classical world (Vol XI).

11 The Ontological Conclusion

The quantization kernel establishes a structural claim:

$$\text{Quantum algebra} = \text{Image}(Q)$$

$$\text{Classical excess} = \text{Null}(Q)$$

The quantum algebra corresponds to the image of a structure-preserving map applied to the classical observable algebra. The classical features that do not survive — commutative products, precise phase-space localization, bracket relations beyond the admissible subalgebra — lie in the null space of Q .

We are careful about what this does and does not assert. It does **not** assert that classical mechanics is ‘correct’ and quantum mechanics is derived from it. It does **not** assert that quantum

mechanics is merely a filtered version of a more fundamental classical reality. What it asserts is that the **relationship** between the two is a kernel — a structure-preserving map with non-trivial null space — and that every known quantization scheme instantiates this relationship through the same three invariants (bracket preservation, non-trivial null space, image/null decomposition), regardless of the specific algebraic mechanism used to realize it.

The uncertainty principle, in this framework, is not a mysterious limitation on measurement. It is the **non-trivial null space** of the quantization kernel: the mathematical statement that Q is not an isomorphism. The minimum phase-space cell $\Delta q \Delta p \geq \hbar/2$ is the kernel’s resolution limit.

The zero-point energy of the harmonic oscillator ($\hbar\omega/2$) is not a property of the classical system or the quantum system alone. It is a property of the **kernel itself** — an artifact of the map. Just as the Hadamard projection in Volume X reveals what was admissible without introducing new content, the quantization kernel reveals which classical observables survive as quantum operators without introducing new physics.

We state this as a formal postulate, extending the observability postulate of Volume X:

Postulate (Quantization as Kernel). The passage from classical to quantum mechanics is a structure-preserving kernel Q whose image is the quantum algebra and whose null space is the classical excess. The three invariants — bracket preservation on an admissible subalgebra, non-trivial null space, and image/null decomposition — are universal across all known quantization schemes.

This completes the POT program through eleven volumes:

- **Volumes I–VI:** Physical laws are properties of admissible kernels.
- **Volume VII:** Renormalization is a projection kernel.
- **Volume VIII:** Conditional reduction of the Yang–Mills mass gap via ITCM.
- **Volume IX:** Yang–Mills vacuum stability as a classical spectral property.
- **Volume X:** All infinities are projection singularities (Hadamard projection).
- **Volume XI:** Quantum mechanics is the image of a quantization kernel.

The chain from Volume I to Volume XI traces a single structural principle: the observable universe is the image of projection kernels. What lies outside the image — divergences, singularities, ordering ambiguities, classical determinism — is not ‘subtracted,’ ‘resolved,’ or ‘lost.’ It was never in the image. It is the null space.

All 26 structural results (19 Z3-verified axioms + 7 computational verifications) are machine-verified in the Kleis formal verification language.

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